

ABSTRACT

This report presents the design and implementation of a desktop-based Student Leave Management System developed using Java Swing, JDBC, and MySQL. The primary objective of the system is to simplify and organize the process of managing student leave applications by providing an efficient and user-friendly interface, thereby reducing manual paperwork and improving record maintenance.

The application supports essential operations such as applying for leave, viewing leave records, updating leave details, and deleting leave applications through a graphical user interface. Each leave record contains information such as student name, reason for leave, leave duration, and approval status. The system is structured using a layered architecture consisting of Data Transfer Object (DTO), Data Access Object (DAO), Service, and User Interface (UI) layers. This separation of concerns improves modularity, maintainability, and scalability of the application.

The DAO layer handles all database interactions using JDBC, enabling secure and efficient communication with the MySQL database. The service layer incorporates business logic and input validation to ensure the consistency and reliability of leave data. The UI layer, developed using Java Swing, provides an intuitive platform for users to interact with the system and perform operations seamlessly.

This project demonstrates the integration of a Java-based graphical interface with a relational database system and highlights the importance of structured software design in developing reliable and maintainable student management applications.